

NISHANTH TATAPUDI

Bangalore, India (Open to relocate to Germany)

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Summary

Certified Software Test Engineer with 8+ years of experience in verification and validation pertaining to the automotive domain. Specialized in HV & LV inverter systems testing for electric vehicles, working across OEM and non-OEM electric mobility platforms, including Mercedes-Benz, Ather, Volvo Trucks, Toyota, and Scania. Delivers safety-critical software validation, defect reduction, and production readiness across cross-functional engineering teams.

Core Skills

- **Functional Safety & Standards:** ISO 26262 (FuSa), ASIL, Safety Validation
- **Verification & Validation / Testing:** System Testing, Regression & Smoke Testing, Test Automation, Debugging & Report Analysis
- **Automotive Protocols:** ISO 11898-CAN, ISO 14229-UDS
- **Process / Compliance:** ASPICE, SYS.2–SYS.3, SYS.5–SYS.4
- **Tools:** CANoe, CANape, NI VeriStand, dSpace, ECU-Test, PCAN View, PUMA, Polarion, JIRA, BWVISU, MATLAB
- **Test Environments:** HIL, PHIL, HV ILB, E-Dyne, Pump Chanc-Dyne
- **ECU/Modules:** HV & LV Inverters, Motor Controllers, VECU (Vehicle ECU), PSS (Power Step Systems), CEPL (Control Panel for External Lighting)

Experience

BorgWarner

July 2023 – Present

Technical Leader – Systems | PDS - Power Electronics Features & Solutions Testing

India & Luxembourg

- Perform HV inverter validation for **Mercedes-Benz** eATS and eATS_MCU (MMA & EVA2M) platform including controls and diagnostics.
- Analyzing System Requirements (SYS.2/SYS.3) and Developing Test Cases (SYS.5/SYS.4) using Black Box Techniques as per ASPICE Compliance.
- Worked on modules like Inverter Controls, PSOL (Position Sensor Offset Learning), FSL2 Flux-based Torque Estimation, FSL2 Power-based Torque Estimation, Electrical Machine Offset Learning, L2-Rotor Temperature Monitoring.
- Perform smoke testing, regression and scenario-based testing on LV/HV and E-Dyne test setups ensuring release readiness and traceability.
- Conduct root cause analysis, defect tracking in Polarion/JIRA and collaborate with calibration teams for SOP.

SPANIDEA Systems Pvt Ltd

October 2020 – July 2023

Senior Software Engineer

Bangalore, India

- Validated LV inverter (Ather Drive Controller) for **Ather EV** platform focusing on performance and thermal stability.
- Tested FMS (Failure Management System) protection features including throttle faults, OCP (SW/HW), motor stall, voltage and thermal protections.
- Verified ECU interactions, UDS diagnostics and Ride/Eco/Sport/Warp mode tuning across 50+ scenarios.
- Executed WOT tests prior to software release for integration and performance test teams.

Global Edge Software Ltd

October 2018 – September 2020

Senior Software Engineer

Bangalore, India

- Performed HIL-based system validation of **Volvo Trucks** VECU over J1939/J1587 multi-ECU networks.
- Validated features including ACC, PTO, ACB, air suspension interlocks and engine speed control logic.
- Analyzed real-time CAN signals and improved ECU integration stability reducing system-level defects.
- Supported **Toyota Magna** PSS and **Scania** BCM programs ensuring CAN compliance and functional robustness.

- Commissioned Energy Management Systems integrating IEC 61850 relays, MFMs and PLC gateways.
- Managed 500+ Modbus RTU devices and easyGen controllers ensuring reliable SCADA data acquisition.
- Validated CAN and Modbus communication using CANalyzer and ModScan for accurate system monitoring.
- Configured SINAMICS G120 drives and pneumatic valve banks for automated plant operations.

Key OEM Projects

Mercedes-Benz eATS – HV Inverter

July 2023 – Present

OEM: Mercedes-Benz | Variants: MMA & EVA2M

India & Luxembourg

- Validated inverter controls including PSOL and torque estimation algorithms.
- Executed flux-based and power-based FSL2 torque validation using HIL and PHIL benches.
- Reduced software defect leakage by 10% through structured regression testing.

ADC – Ather Drive Controller (LV Inverter)

October 2020 – July 2023

OEM: Ather Energy

Bangalore, India

- Validated traction controller for performance, range, drivability and thermal stability.
- Performed throttle error, motor stall and protection logic validation.
- Improved controller robustness reducing field fault incidents by <5%.

Tea2BP VTNA/MACK – VECU Validation

October 2018 – September 2020

OEM: Volvo Trucks

Bangalore, India

- Conducted system verification of VECU features across multi-ECU J1939 network.
- Validated gateway, speed limiting and external control interactions via HIL.
- Improved system stability reducing cross-ECU defects by 15%.

Education

JNTU Anantapur

May 2010 – May 2013

B.E. Electrical & Electronics Engineering

Kavali, India

Sri Venkateswara University

April 2006 – May 2009

Diploma in Electrical & Electronics Engineering

Proddatur, India

Certifications

- TÜV Rheinland — High Voltage Electrical Safety (March 2024)
- ISTQB — Certified Software Tester (December 2023)
- Post Graduate Diploma in Embedded Systems Design & Development (March 2018)
- SIMATIC S7 1200/1500 with TIA V13 (June 2016)
- Advanced Training in Automation and Control Engineering (March 2015)

Languages

- English — Full Professional Proficiency (C1)
- German — Intermediate Working Proficiency (B1)